

Weighted utility

Model — human; produced by literature.

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.
- **Betweenness.** For all $p, q \in \Delta(X)$ and $\lambda \in (0,1)$: if $p \succ q$ then $p \succ \lambda p + (1-\lambda)q \succ q$, and if $p \sim q$ then $p \sim \lambda p + (1-\lambda)q$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.

0.2 Construction

Weighted utility. Fix $u : X \rightarrow \mathbb{R}$ and a weight $w : X \rightarrow (0, \infty)$ that is not constant, and let $V(p) = \sum p(x)w(x)u(x) / \sum p(x)w(x)$; put $p \succcurlyeq q$ iff $V(p) \geq V(q)$. V is continuous on $\Delta(X)$, so $\succcurlyeq$ is a weak order with closed graph. Its indifference sets are the intersections of $\Delta(X)$ with hyperplanes $\{p : \sum p(x)w(x)(u(x) - v) = 0\}$, which are straight, so betweenness holds. Since w is not constant these hyperplanes are not parallel, so $\succcurlyeq$ is not an expected-utility order and (being a continuous weak order) must violate independence.

0.3 Sources

- Chew (1983), A generalization of the quasilinear mean with applications to the measurement of income inequality and decision theory resolving the Allais paradox, *Econometrica* 51
- Dekel (1986), An axiomatic characterization of preferences under uncertainty: weakening the independence axiom, *JET* 40

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